ASCOMP 2026 POSTER

Min-Marginal Power Indices for Multi-Objective Feature Selection

SCAN ME

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01 PRINCIPLE & TOOLKIT

A coalition witness reveals harmful additions: acceptable utility becomes unacceptable.

N: analyzed features; $n = |N|, \quad U_{\text{val}}(N) > 0, \quad 0 < \alpha \le 1.$

$$v_\alpha(S) = \mathbf{1}[U_{\text{val}}(S) \ge \alpha U_{\text{val}}(N)]$$

$$\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$$

$$\text{MinPI}_i = \text{MinSS}_i = \text{MinBz}_i = \min_{S \subseteq N \setminus \{i\}} \Delta_i(S)$$

Exact MinPI labels:
-1 **harmful:** some coalition flips $1 \to 0$.
0 **neutral:** all marginals ≥ 0; at least one is 0.
$+1$ **decisive:** every coalition flips $0 \to 1$.

MinSS / MinBz: minimum Shapley-Shubik / Banzhaf.
Both minima cover all coalitions; exact uses at most 2^n utilities.

CORE TOOLS

MI (mutual information) ranks feature-target dependence.
A **maximum-weight absolute-correlation tree** represents pairwise association and guides Algorithm B.

XGBoost: 100 rounds, depth 4, learning rate 0.1; CUDA trees. Fit coalition models on training data.

Training-only preprocessing: variance filtering → MI selection ($\le$100 features) → standardization.

Utility: ROC AUC / one-vs-rest AUC / R^2.
$U_{\text{val}}(S)$: validation utility of train-fitted subset S.
Empty coalition: training class priors / response mean.

MATCHED-COUNT CONTROLS

All controls delete the accepted MinPI count k.
SHAP-bottom: lowest mean |TreeSHAP| on train.
MinShap / Banzhaf-bottom: lowest minimum / mean continuous utility marginal from shared coalitions.
Random: 30 matched-count trials per seed.

Quota sweep (seed 42): $\alpha \in \{0.85, 0.90, 0.95, 0.99\}$.

02 ALGORITHM FLOW

Train 64% → validation 16% → test 20%
Fit on train; compute the threshold game on validation.

Algorithm A: size-uniform sampling

Draw s uniformly from $\{0, \dots, n-1\}$; then draw an s-feature subset uniformly from $N \setminus \{i\}$.

Algorithm B: MI-correlation guidance

B_i: up to 10 tree neighbors ranked by absolute correlation; pad from the top-MI half of $N \setminus \{i\}$ toward $\min\{20, \lfloor n/3 \rfloor\}$.
Each draw: reset $S \leftarrow B_i$; drop each independently ($p = 0.3$).
Draw a uniformly from $\{0, \dots, \min(5, n-1)\}$;
add $\min\{a, n - 1 - |S|\}$ absent features in random order.

Initialize $m \leftarrow +1$; update $m \leftarrow \min\{m, \Delta_i(S)\}$.
Stop at -1 with a witness; else return m after K draws.
$K \in \{2, 5, 10, 25\}$ per feature per estimator.
ISOLET: $K \in \{2, 5, 10\}$; removal uses the largest K.

HYBRID DISPATCH

$n \le 20$: exact | $20 < n \le 75$: B* | $n > 75$: $\min(A, B)$
*Use A if no correlation tree is available; min is featurewise.

Validate joint removal

$$H = \{i \in N : \widehat{\text{MinPI}}_i = -1\}$$

$$U_{\text{val}}(N \setminus H) \ge U_{\text{val}}(N) - \varepsilon$$

$\varepsilon = 0.005$ (classification); 0.01 (regression).
Remove $R = H$ on **PASS**; $R = \emptyset$ on **FAIL**.
Test $S_{\text{ret}} = N \setminus R$ once; controls delete $k = |R|$.

Optional MinPI-prefiltered NSGA-II

Non-dominated sorting genetic algorithm II; minimize:

$$\left(\frac{|S|}{n},\ 1 - \text{clip}(U_{\text{val}}(S), 0, 1) \right)$$

n is shared: analyzed dimension before prefiltering.
population 40; 800 evaluations per variant.
HV reference (1.1, 1.1); select fronts on validation.
Evaluate the selected fronts on held-out test data.

03 EXPERIMENTAL EVIDENCE

6 datasets × 5 seeds

Seeds: 42, 123, 456, 789, 1024 | default α = 0.95

12.53%	-0.000313
mean feature removal	mean held-out ΔU

$\Delta U = U_{\text{test}}(S_{\text{ret}}) - U_{\text{test}}(N)$; five-seed mean ± SD.
Analyzed-feature removal; bars: 0-80%, whiskers: SD.

DATASET	REMOVED (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	+0.00469 ± 0.00497
Abalone	27.50 ± 25.62	-0.00565 ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	-0.00078 ± 0.00335
Spambase	0.72 ± 0.99	-0.00003 ± 0.00007
Gisette	0.60 ± 0.89	-0.00011 ± 0.00016
ISOLET	0.00 ± 0.00	0.00000 ± 0.00000

MATCHED-COUNT REMOVAL: MinPI vs random
Madelon: ΔU **+0.0047 vs -0.0732**
Abalone: ΔU **-0.0056 vs -0.0370**

NSGA-II: held-out hypervolume

$\Delta\text{HV} = \text{HV}_{\text{prefiltered}} - \text{HV}_{\text{full}}$.
Mean +0.01584 (Holm $p = 0.4375$); Madelon +0.09515.
Positive dataset-averaged change in all five seeds.

ESTIMATOR ACCURACY

Abalone: mean harmful-set F_1 vs exact; 8 features, 5 seeds.
Hybrid uses exact enumeration: $F_1 = 1$ at every K.

K	2	5	10	25
A	0.100	0.150	0.230	0.443
B	0.000	0.080	0.133	0.313